1

(10 pt) The data in this study contains information about crime in States across the United States in 2005. The variables include the number of violent crimes per 100,000 people (Violent), the number of murders per 100,000 people (Murder), the percentage of people living in metropolitan areas (Metro), the percentage of the White population (White), the percentage of high school graduates (High School), and the percentage of people living below the poverty level (Poverty).

(i)

(2 pt) Fit a Poisson GLM to predict the expected murder rate based on the predictors: Poverty, Metro, and White.

```r
crime = read.table("data/Statewide_crime.txt", header = T)

fit = glm(Murder~Poverty+Metro+White, data=crime, family=poisson(link="log"))

summary(fit)
```

```
Call:
glm(formula = Murder ~ Poverty + Metro + White, family = poisson(link = "log"),
    data = crime)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.508089   0.681361   0.746 0.455850
Poverty      0.121155   0.017876   6.777 1.22e-11 ***
Metro        0.014394   0.004334   3.321 0.000896 ***
White       -0.014343   0.004288  -3.345 0.000824 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 178.216  on 50  degrees of freedom
Residual deviance:  38.984  on 47  degrees of freedom
AIC: 217.33

Number of Fisher Scoring iterations: 5
```

(ii)

(2 pt) Another investigator suspects that the murder rate is proportional to the number of violent crimes. Re-fit the Poisson GLM in (i) and include Violent as an offset.

```r
fit2 = glm(Murder~Poverty+Metro+White+Violent, data=crime, family=poisson(link="log"), offse
summary(fit2)
```

```
Call:
glm(formula = Murder ~ Poverty + Metro + White + Violent, family = poisson(link = "log"),
    data = crime, offset = log(Violent))

Coefficients:
              Estimate Std. Error z value Pr(>|z|)    
(Intercept) -4.1701028  0.7450396  -5.597 2.18e-08 ***
Poverty      0.0618450  0.0238030   2.598  0.00937 ** 
Metro        0.0005636  0.0054097   0.104  0.91703    
White       -0.0100210  0.0055607  -1.802  0.07153 .  
Violent     -0.0002191  0.0003183  -0.688  0.49124    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 53.002  on 50  degrees of freedom
Residual deviance: 25.803  on 46  degrees of freedom
AIC: 206.15

Number of Fisher Scoring iterations: 4
```

(iii)

(2 pt) Are the results in (i) and (ii) the same? What did you observe?

> No the results are not the same. All predictors were highly significant in the first fit, but in the second model Metro is no longer statistically significant and the p-value for white is ≈ 0.072 so it may not be considered significant at $\alpha = 0.05$. Even for the Poverty predictor, while the predictor is still statistically significant, the coefficient decreased from 0.121 to 0.062 in the 2nd model.

(iv)

(1 pt) Calculate the ratio of the residual deviance to the degrees of freedom (residual deviance/df) for the model in (ii).

> According to the summary output in (ii), the residual deviance is 25.803 and the df is 46 so the ratio is 0.5609347826

(v)

(1 pt) Do you believe there is over-dispersion in the model (ii) based on your result in (iv)? [Note values of residual deviance/df larger than 1 suggest overdispersion is present].

> The disperion parameter calculated in part (iv) suggest there is not over-dispersion because the parameter is less than 1.

(vi)

(2pt) Re-fit the model in (ii) using a quasi-Poisson GLM to control for overdispersion. Is your conclusion from (ii) different from (vi)?

```r
fit_quasi = glm(Murder~Poverty+Metro+White+Violent, data=crime, family=quasipoisson(link="log

summary(fit_quasi)
```

```
Call:
glm(formula = Murder ~ Poverty + Metro + White + Violent, family = quasipoisson(link = "log")
    data = crime, offset = log(Violent))

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -4.1701028  0.5586588  -7.464 1.84e-09 ***
Poverty      0.0618450  0.0178484   3.465  0.00116 **
Metro        0.0005636  0.0040564   0.139  0.89011
White       -0.0100210  0.0041696  -2.403  0.02033 *
Violent     -0.0002191  0.0002387  -0.918  0.36342
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for quasipoisson family taken to be 0.5622565)
```

```
    Null deviance: 53.002  on 50  degrees of freedom
Residual deviance: 25.803  on 46  degrees of freedom
AIC: NA

Number of Fisher Scoring iterations: 4
```

Here, the β's are all the same with the standard error decreasing for the β's thus resulting in increased statistical significance for the White predictor. The over-dispersion parameter calculated in part (vi) remains the same for this model.